

Debojyoti Ghosh

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RESEARCH INTERESTS

- Simulation data and solvers for scientific machine learning and reduced-order modeling; exploratory work on learned denoisers for particle-in-cell data.
- Numerical methods for PDE-based models: high-order finite-difference and finite-volume schemes, implicit and semi-implicit time integration.
- Domain discretization: overset meshes, mapped multi-block grids, immersed boundary methods.
- Particle-based and agent-based methods.
- High-performance computing: design and implementation of scalable algorithms on DOE leadership-class platforms.
- *Applications*: fusion and high-energy-density plasmas, atmospheric and cloud physics, compressible and incompressible aerodynamics, epidemiology.

PROFESSIONAL EXPERIENCE

Computational Scientist

Feb 2018 – present

Center for Applied Scientific Computing, Lawrence Livermore National Laboratory

- Machine learning for simulation: conditioned neural denoisers for implicit particle-in-cell methods; reduced-order models for plasma dynamics and cloud microphysics.
- Particle-based methods: super-droplet cloud microphysics, including ice processes, in a performance-portable numerical weather prediction code; implicit particle-in-cell algorithms for dense plasmas.
- Agent-based models: agent interaction fidelity, disease spread, and capacity-limited hospital care in an exascale-capable epidemiological code.
- High-order discretization and implicit-explicit time integration for magnetically confined plasmas, atmospheric flows, and gas optics.
- Multifluid models for interpenetrating plasmas in inertial confinement fusion.

Postdoctoral Research Staff Member

Oct 2015 – Feb 2018

Center for Applied Scientific Computing, Lawrence Livermore National Laboratory

- Implicit-explicit time integration for continuum kinetic simulation of collisional magnetized plasmas.
- High-order finite-volume algorithms for tokamak edge plasma simulation.
- Multirate semi-implicit integrators for adaptive-mesh atmospheric flow solvers.

Postdoctoral Appointee

Feb 2013 – Oct 2015

Mathematics and Computer Science Division, Argonne National Laboratory

Fellow

Mar 2015 – Oct 2015

Computation Institute, The University of Chicago

- Implemented multi-stage implicit, explicit, and implicit-explicit Runge-Kutta schemes, and general linear methods with global error estimation, in PETSc.
- Applied high-order time integrators to production physics codes, including numerical weather prediction.
- Scalable implementation of nonlinear compact finite-difference schemes on DOE leadership-class platforms.
- Well-balanced, conservative finite-difference methods for limited-area atmospheric flows.
- Probability density function methods for power systems with uncertain input.

Graduate Research Assistant

Aug 2008 – Jan 2013

Alfred Gessow Rotorcraft Center, Aerospace Engineering, University of Maryland

- Derived, analysed, and implemented the CRWENO family of high-resolution non-oscillatory compact reconstruction schemes.
- Developed a three-dimensional Cartesian incompressible Navier-Stokes solver with immersed boundaries for high-Reynolds-

number flows.

- Applied these methods to direct numerical simulation of compressible turbulence, vortex-ring and wall interaction, and rotorcraft wake modeling.

Research Assistant

2003 – Jun 2006

Department of Aerospace Engineering, Indian Institute of Technology Bombay

- High-order non-oscillatory schemes for ideal magnetohydrodynamics, including an analysis of non-convexity effects on convergence.
- Finite-volume time-domain solution of Maxwell's equations to predict the radar cross-section of low-observable aircraft.

Summer Intern

May – Aug 2004

Infotech Enterprises, Pratt & Whitney Canada Center of Excellence, Bangalore

- Computational analysis of an effusion-cooled plate representing a gas turbine combustion chamber casing.

EDUCATION

Doctor of Philosophy, Applied Mathematics and Statistics, and Scientific Computation

Jan 2013

University of Maryland, College Park • Adviser: James D. Baeder

Master of Technology and Bachelor of Technology (dual degree), Aerospace Engineering

Jul 2006

Indian Institute of Technology Bombay • Adviser: Avijit Chatterjee

TRAINING PROGRAMS

- Generative AI, industrial short course, Institute for Pure and Applied Mathematics, UC Los Angeles, March 2026.
- End-to-End AI for Science Bootcamp, NERSC, December 2025.
- Reinforcement Learning course, 2024.
- Computational Machine Learning for Scientists and Engineers, ECE Continuum, University of Michigan, June 2021.
- Argonne Training Program on Extreme-Scale Computing (ATPESC), August 2014.

TECHNICAL SKILLS

Machine learning	PyTorch, distributed data-parallel training, U-Net and encoder–decoder architectures, diffusion models, attention, autoencoders, physics-informed objectives, reduced-order modeling
Languages	C++, C, Python, CUDA, MATLAB
Parallel	MPI, OpenMP, CUDA, AMReX, multi-GPU training
Libraries	PETSc, SUNDIALS, Chombo, hypre, FFTW, NumPy, yt
Tooling	Git, CMake, GNU Autotools, Spack, pytest, Jupyter, Blitz $\text{\LaTeX}$
AI-assisted dev	Claude Code, OpenAI Codex, Cursor
Visualization	VisIt, Matplotlib, Tecplot

SCIENTIFIC SOFTWARE

- **HyPar** (developer) — conservative finite-difference solver for n -dimensional hyperbolic-parabolic PDEs.
- **EUCLID** (developer) — high-order conservative finite-difference code for multifluid plasma simulation.
- **TridiagLU** (developer) — parallel solvers for distributed tridiagonal systems.
- **particle-denoise** (developer) — PyTorch denoiser for particle-in-cell simulation moments. Early-stage, unpublished.
- **ERF** (lead developer of the Super-Droplet Method; 53 merged pull requests) — performance-portable numerical weather prediction code. Particle-based cloud microphysics including ice processes, AMR support, terrain-following particle advection, and GPU portability.
- **ExaEpi** (key developer; 31 merged pull requests) — exascale-capable agent-based epidemiological model. Multi-disease tracking with co-immunity and co-susceptibility, capacity-limited hospitalization, and the agent-interaction kernels.
- **WarpX** (solver and preconditioner developer; 7 merged pull requests) — exascale particle-in-cell code. Curl-curl preconditioner, sparse-matrix Jacobian assembly, and the PETSc SNES/KSP interface for implicit electromagnetic PIC.

- **PETSc** (contributor, 57 commits, 2013–2015) — the **TSGLEE** general linear methods with global error estimation; the **MATKAIJ** matrix format for implicit Runge-Kutta stage systems; Dormand-Prince and Bogacki-Shampine embedded Runge-Kutta schemes; extrapolated stage predictors for TSARKIMEX. All still in the library. <https://gitlab.com/petsc/petsc/-/commits/main?author=Debojyoti+Ghosh>
- **COGENT** (contributor) — high-order continuum gyrokinetic code for tokamak edge simulation.
- **PICNIC** (contributor) — particle-in-cell code with Coulomb collisions.
- **PIAFS** (co-developer) — high-order nonlinear hydrodynamics with photochemistry for gaseous optics, including the CUDA implementation. <https://github.com/LLNL/piafs>
- **NUMA** (contributor) — massively parallel numerical weather prediction code.

JOURNAL ARTICLES (29) — 1000+ CITATIONS, H-INDEX 16, I10-INDEX 22 (GOOGLE SCHOLAR, OCTOBER 2026)

- de Jong, E. K., Gunawardena, N., Katona, J. E., Beydoun, H., **Ghosh, D.**, Caldwell, P., *Data-Driven Reduced Order Modeling for Warm Rain Microphysics*, Journal of Geophysical Research: Machine Learning and Computation, 3 (3), 2026, e2025JH001103, doi:10.1029/2025JH001103.
- Tsai, P.-H., Chung, S. W., **Ghosh, D.**, Loffeld, J., Choi, Y., Belof, J. L., *Local reduced-order modeling for electrostatic plasmas by physics-informed solution manifold decomposition*, Computer Physics Communications, 322, 2026, 110039, doi:10.1016/j.cpc.2026.110039.
- Michel, P., Oudin, A., Rajesh, H., Ou, K., Chakraborty, D., Cao, S., Kur, E., Lancia, L., **Ghosh, D.**, Riconda, C., Wurtele, J. S., Edwards, M. R., *Entropy-mode-driven gas optics*, Physical Review Applied, 25, 2026, 024037, doi:10.1103/pnhn-w6tl.
- Salian, L. V., Samala, R., **Ghosh, D.**, *Central compact finite-difference scheme with high spectral resolution for KdV equation*, Numerical Methods for Partial Differential Equations, 42 (1), 2026, e70060, doi:10.1002/num.70060.
- Dorf, M., Dorr, M., Geyko, V., **Ghosh, D.**, Umansky, M., Angus, J., *Semi-implicit continuum kinetic modeling of weakly collisional parallel transport in a magnetic mirror*, Physics of Plasmas, 32, 2025, 103903, doi:10.1063/5.0293154.
- van de Wetering, J., Angus, J. R., Farmer, W., Geyko, V., **Ghosh, D.**, Grote, D., Weber, C., Zimmerman, G., *Particle-in-cell simulations of burning inertial confinement fusion capsule implosions*, Physical Review E, 112, 2025, 045207, doi:10.1103/zwjx-jbxi.
- Dorf, M., Dorr, M., **Ghosh, D.**, Umansky, M., Soukhanovskii, V., *Implicit full-F simulations of neoclassical ion transport*, Physics of Plasmas, 32, 2025, 083904, doi:10.1063/5.0280028.
- Oudin, A., **Ghosh, D.**, Riconda, C., Lancia, L., Kur, E., Ou, K., Perez-Ramirez, V. M., Lee, J., Edwards, M. R., Michel, P., *PIAFS: A 2D nonlinear hydrodynamics code to model gaseous optics*, Physics of Plasmas, 32, 2025, 072714, doi:10.1063/5.0268318.
- Geyko, V. I., Joseph, I., Dorf, M. A., **Ghosh, D.**, Dorr, M. R., *Drift kinetic electrostatic simulations of the edge localized mode heat pulse*, Physics of Plasmas, 31, 2024, 123903, doi:10.1063/5.0230913.
- Angus, J. R., Farmer, W., Friedman, A., Geyko, V., **Ghosh, D.**, Grote, D., Larson, D., Link, A., *An implicit particle code with exact energy and charge conservation for studies of dense plasmas in axisymmetric geometries*, Journal of Computational Physics, 519, 2024, 113427, doi:10.1016/j.jcp.2024.113427.
- Dorf, M., Dorr, M., **Ghosh, D.**, *Development of an implicit electromagnetic capability for a hybrid gyrokinetic ion-fluid electron model*, Contributions to Plasma Physics, 64 (7-8), 2024, doi:10.1002/ctpp.202300114.
- Bonneville, C., Choi, Y., **Ghosh, D.**, Belof, J. L., *GPLaSDI: Gaussian Process-based interpretable Latent Space Dynamics Identification through deep autoencoder*, Computer Methods in Applied Mechanics and Engineering, 418 (A), 2024, 116535, doi:10.1016/j.cma.2023.116535.
- Angus, J. R., Farmer, W., Friedman, A., **Ghosh, D.**, Grote, D., Larson, D., Link, A., *An implicit particle code with exact energy and charge conservation for electromagnetic studies of dense plasmas*, Journal of Computational Physics, 491, 2023, 112383, doi:10.1016/j.jcp.2023.112383.
- Kim, Y., **Ghosh, D.**, Constantinescu, E. M., Balakrishnan, R., *GPU-accelerated DNS of compressible turbulent flows*, Computers and Fluids, 251, 2023, 105744, doi:10.1016/j.compfluid.2022.105744.
- Angus, J. R., Link, A., Friedman, A., **Ghosh, D.**, Johnson, J. D., *On numerical energy conservation for an implicit particle-in-cell method coupled with a binary Monte-Carlo algorithm for Coulomb collisions*, Journal of Computational Physics, 456, 2022, doi:10.1016/j.jcp.2022.111030.
- Chapman T., Winjum B. J., Berger, R. L., Dimits, A., Banks, J. W., Brunner, S., Joseph, I., **Ghosh, D.**, *Nonlinear kinetic simulation study of the ion-ion streaming instability in single- and multi-ion species plasmas*, Physics of Plasmas, 28, 022105, 2021, doi:10.1063/5.0015302.
- Ghosh, D.**, Chapman, T. D., Berger, R. L., Dimits, A., Banks, J. W., *A Multispecies, Multifluid Model for Laser-Induced Counterstreaming Plasma Simulations*, Computers and Fluids, 186, 2019, 38-57, doi:10.1016/j.compfluid.2019.04.012.
- Dimits, A. M., Banks, J. W., Berger, R. L., Brunner, S., Chapman, T. D., Copeland, D., **Ghosh, D.**, Arrighi, W. J., Hittinger, J., Joseph, I., *Linearized Coulomb Collision Operator for Simulation of Interpenetrating Plasma Streams*, IEEE Transactions on Plasma Science, 47 (5), 2019, 2074-2080, doi:10.1109/TPS.2019.2897790.
- Ghosh, D.**, Dorf, M. A., Dorr, M. R., Hittinger, J., *Kinetic Simulation of Collisional Magnetized Plasmas with Semi-Implicit Time Integration*,

- Journal of Scientific Computing, 77 (2), 2018, 819-849, doi:10.1007/s10915-018-0726-6.
- Dorr, M., Colella, P., Dorf, M., **Ghosh, D.**, Hittinger, J., Schwartz, P. O., *High-order Discretization of a Gyrokinetic Vlasov Model in Edge Plasma Geometry*, Journal of Computational Physics, 373, 2018, 605-630, doi:10.1016/j.jcp.2018.07.008.
- Dorf, M., Dor, M., Hittinger, J., Lee, W., **Ghosh, D.**, *High-order finite-volume modeling of drift waves*, Journal of Computational Physics, 373, 2018, 446-454, doi:10.1016/j.jcp.2018.07.009.
- Ghosh, D.**, Constantinescu, E. M., *Well-Balanced, Conservative Finite-Difference Algorithm for Atmospheric Flows*, AIAA Journal, 54 (4), 2016, 1370-1385, doi:10.2514/1.J054580.
- Ghosh, D.**, Constantinescu, E. M., *Semi-Implicit Time Integration of Atmospheric Flows with Characteristic-Based Flux Partitioning*, SIAM Journal on Scientific Computing, 38 (3), 2016, A1848-A1875, doi:10.1137/15M1044369.
- Wang, P., Barajas-Solano, D. A., Constantinescu, E. M., Abhyankar, S., **Ghosh, D.**, Smith, B. F., Huang, Z., Tartakovsky, A. M., *Probabilistic Density Function Method for Stochastic ODEs of Power Systems with Uncertain Power Input*, SIAM/ASA Journal on Uncertainty Quantification, 3 (1), 2015, 873-896, doi:10.1137/130940050.
- Ghosh, D.**, Constantinescu, E. M., Brown, J., *Efficient Implementation of Nonlinear Compact Schemes on Massively Parallel Platforms*, SIAM Journal on Scientific Computing, 37 (3), 2015, C354-C383, doi:10.1137/140989261.
- Ghosh, D.**, Baeder, J. D., *Weighted Non-Linear Compact Schemes for the Direct Numerical Simulation of Compressible, Turbulent Flows*, Journal of Scientific Computing, 61 (1), 2014, 61-89, doi:10.1007/s10915-014-9818-0.
- Ghosh, D.**, Medida, S., Baeder, J. D., *Application of Compact-Reconstruction Weighted Essentially Nonoscillatory Schemes to Compressible Aerodynamic Flows*, AIAA Journal, 52 (9), 2014, 1858-1870, doi:10.2514/1.J052654.
- Ghosh, D.**, Baeder, J. D., *A High-Order Accurate Incompressible Navier Stokes Algorithm for Vortex Ring Interactions with Solid Wall*, AIAA Journal, 50 (11), 2012, 2408-2422, doi:10.2514/1.J051537.
- Ghosh, D.**, Baeder, J. D., *Compact Reconstruction Schemes with Weighted ENO Limiting for Hyperbolic Conservation Laws*, SIAM Journal on Scientific Computing, 34 (3), 2012, A1678-A1706, doi:10.1137/110857659.

BOOK CHAPTERS (1)

- Ghosh, D.**, Constantinescu, E. M., *Nonlinear Compact Finite-Difference Schemes with Semi-Implicit Time Stepping*, in Spectral and High Order Methods for Partial Differential Equations ICOSAHOM 2014, Springer Lecture Notes in Computational Science and Engineering, Volume 106, 2015, 237-245, doi:10.1007/978-3-319-19800-2_20.

PEER-REVIEWED CONFERENCE PAPERS (12)

- Tsai, P.-H., Chung, S. W., **Ghosh, D.**, Loffeld, J., Choi, Y., Belof, J. L., *Accelerating Kinetic Simulations of Electrostatic Plasmas with Reduced-Order Modeling*, Workshop on Machine Learning and the Physical Sciences, 37th conference on Neural Information Processing Systems (NeurIPS), December 10 - 16, 2023, New Orleans, LA.
- Bonneville, C., Choi, Y., **Ghosh, D.**, Belof, J. L., *Data-Driven Autoencoder Numerical Solver with Uncertainty Quantification for Fast Physical Simulations*, Workshop on Machine Learning and the Physical Sciences, 37th conference on Neural Information Processing Systems (NeurIPS), December 10 - 16, 2023, New Orleans, LA.
- Dorf, M., Dor, M., **Ghosh, D.**, Umansky, M., *A hybrid gyrokinetic ion – fluid electron model for edge plasma simulations*, 29th IAEA Fusion Energy Conference, October 16 - 21, 2023, London, UK.
- Christopher, J., Davis, A., Gao, X., **Ghosh, D.**, Dorf, M., Dor, M., Ricketson, L., *On Preconditioning for the Two-Field Neutral Model*, AIAA Scitech 2022 Forum, January 3 - 7, 2022, San Diego, CA, doi:10.2514/6.2022-1108.
- Munafo, A., Vogiatzis, K., **Ghosh, D.**, Vedula, P., Panesi, M., Josyula, E., *High-Order Techniques for Multi-Component Turbulent Non-Equilibrium Hypersonic Flows*, AIAA Scitech 2020 Forum, January 6 - 10, 2020, Orlando, FL, doi:10.2514/6.2020-2193.
- Dorf, M. A., Dor, M., **Ghosh, D.**, Hittinger, J., Umansky, M. V., Angus, J., Pankin, A., *Simulation of Cross-Separatrix Edge Plasma Transport with the Continuum Gyrokinetic Code COGENT*, 27th IAEA Fusion Energy Conference, October 22 - 27, 2019, Gandhinagar, Gujarat, India.
- Guhur, P.-L., Constantinescu, E. M., **Ghosh, D.**, Peterka, T., Cappello, F., *Detection of Silent Data Corruption in Adaptive Numerical Integration Solvers*, IEEE Cluster Conference, September 5-8, 2017, Honolulu, HI, doi:10.1109/CLUSTER.2017.13.
- Ghosh, D.**, Dorf, M., Hittinger, J., Dor, M., *Implicit-Explicit Time Integration for the Vlasov-Fokker-Planck Equations*, AIAA Paper 2017-4016, 48th AIAA Plasmadynamics and Lasers Conference, June 5-9, 2017, Denver, CO, doi:10.2514/6.2017-4016.
- Ghosh, D.**, Constantinescu, E. M., *Well-Balanced Formulation of Gravitational Source Terms for Conservative Finite-Difference Atmospheric Flow Solvers*, AIAA Paper 2015-2889, 7th AIAA Atmospheric and Space Environments Conference, June 22-26, 2015, Dallas, TX, doi:10.2514/6.2015-2889.
- Ghosh, D.**, Medida, S., Baeder, J. D., *Compact-Reconstruction Weighted Essentially Non-Oscillatory Schemes for Unsteady Navier-Stokes*

Equations, AIAA Paper 2012-2832, 42nd AIAA Fluid Dynamics Conference and Exhibit, June 25-28, 2012, New Orleans, LA, doi:10.2514/6.2012-2832.

Ghosh, D., Baeder, J. D., *Numerical Simulation of Vortex Ring Interactions with Solid Wall*, 49th AIAA Aerospace Sciences Meeting, Jan 4-7, 2011, Orlando, FL, doi:10.2514/6.2011-675.

Ghosh, D., Baeder, J. D., *A High Order Conservative Upwind Algorithm for the Incompressible Navier Stokes Equations*, AIAA Paper 2010-5030, 40th AIAA Fluid Dynamics Conference and Exhibit, June 28 – July 1, 2010, Chicago, IL, doi:10.2514/6.2010-5030.

TECHNICAL REPORTS (2)

Terasaki, G., **Ghosh, D.**, Navid, A., *Improved School and Workplace Interactions in ExaEpi, an Agent-Based Epidemiological Model*, Lawrence Livermore National Laboratory, Livermore, CA, 2025.

Ghosh, D., Constantinescu, E. M., Brown, J., *Scalable Nonlinear Compact Schemes*, Technical Memorandum, ANL/MCS-TM-340, Argonne National Laboratory, April 2014.

THESES (3)

Compact-Reconstruction Weighted Essentially Non-Oscillatory Schemes for Hyperbolic Conservation Laws, Ph. D. Thesis, University of Maryland, College Park, January 2013.

Higher Order Non-Oscillatory Schemes in Ideal Magnetohydrodynamics, Masters Thesis, Indian Institute of Technology Bombay, July 2006.

Finite Volume Time Domain Solutions of Maxwell's Equations, Bachelors Seminar, Indian Institute of Technology Bombay, March 2004.

INVITED AND MINISYMPOSIUM TALKS (35)

Ghosh, D., Beydoun, H., McGuffin, D., Gardner, D. J., Lundquist, K., *A scalable particle-based microphysics model for atmospheric flow simulations*, 18th U.S. National Congress on Computational Mechanics (USNCCM), Minisymposium on Computational fluid dynamics (CFD) and fluid-structure interaction (FSI): method development and applications, July 20 - 24, 2025, Chicago, IL.

Angus, J. R., Farmer, W., Friedman, A., Fu, Y., Geyko, V., **Ghosh, D.**, Grote, D., Larson, D., Ludwig, J., Van de Wetering, J., Zimmerman, G., Link, A., *An implicit particle code for fully kinetic simulations of dynamically compressing plasmas*, IEEE Pulsed Power and Plasma Science Conference (PPPS 2025), Session on Computational Plasmas, June 15 - 20, 2025, Berlin, Germany.

Ghosh, D., Beydoun, H., McGuffin, D., Gardner, D. J., Lundquist, K., *Super Droplet Method for Lagrangian Microphysics Simulations in ERF*, SIAM Conference on Computational Science and Engineering, Minisymposium on Energy Research and Forecasting (ERF): A New Performance-Portable Atmospheric Modeling Code, March 3 - 7, 2025, Fort Worth, TX.

Myers, A., Nugent, P., Singhal, A., Acquesta, E., **Ghosh, D.**, Keilbart, N., Nguyen, T., *ExaEpi: An Exascale-Capable Agent-Based Model for Epidemiology*, SIAM Conference on Computational Science and Engineering, Minisymposium on Predictive Disease Modeling and Simulations for Decision-making, March 3 - 7, 2025, Fort Worth, TX.

Tsai, P.-H., Chung, S. W., **Ghosh, D.**, Loffeld, J., Choi, Y., Belof, J., *Accelerating Kinetic Simulations of Electrostatic Plasmas with Local Reduced-Order Modeling*, SIAM Conference on Computational Science and Engineering, Minisymposium on Scientific Machine Learning at Scale, March 3 - 7, 2025, Fort Worth, TX.

Dorf, M., Dorr, M., **Ghosh, D.**, *A Semi-Implicit Gyrokinetic Ion – Fluid Electron Hybrid Model for Edge Plasma Simulations*, Platform for Advanced Scientific Computing (PASC), Minisymposium on High Performance Computing for Magnetic Fusion Applications - Part II, June 3 - 5, 2024, Zurich, Switzerland.

Ghosh, D., Dorf, M., Dorr, M., *Magnetized Plasma Simulations with High-Order Implicit-Explicit Time Integrators*, 9th European Congress on Computational Methods in Applied Sciences and Engineering (ECCOMAS), Minisymposium on Computational Kinetic Theory II, June 3 - 7, 2024, Lisbon, Portugal.

Tsai, P.-H., Chung, S. W., **Ghosh, D.**, Loffeld, J., Choi, Y., Belof, J., *Projection-Based Reduced Order Model for Accelerating Kinetic Simulations of Electrostatic Plasmas*, SIAM Conference on Parallel Processing for Scientific Computing (PP), Minisymposium on HPC Algorithms for Kinetic Equations for Plasmas, March 5 - 8, 2024, Baltimore, MD.

Choi, Y., Copeland, D., Cheung, S. W., Kim, Y., He, X., Fries, W., Bonneville, C., **Ghosh, D.**, Shin, Y., Park, J. S., Belof, J., Afeyan, B., *Physics-Guided Latent Space Dynamics Identification in Model Order Reduction*, 17th U.S. National Congress on Computational Mechanics (USNCCM), Minisymposium on Model Order Reduction for Parameterized Continuum Mechanics, July 23 - 27, 2023, Albuquerque, NM.

Ghosh, D., Dorf, M., Dorr, M., Ricketson, L., *A Multiphysics Implicit-Explicit Time Integration Algorithm for Simulating Tokamak-Edge Plasma Dynamics*, 17th U.S. National Congress on Computational Mechanics (USNCCM), Minisymposium on Advances in Numerical Methods for Solution of PDEs, July 23 - 27, 2023, Albuquerque, NM.

Bonneville, C., He, X., Chen, J.-S., Fries, W., Cheung, S. W., **Ghosh, D.**, Belof, J., Springer, K., Choi, Y., *Certified and Parameterized Latent*

- Space Dynamics Identification for Time Dependent Image Data*, AAAI Spring Symposium on Computational Approaches to Scientific Discovery, March 27 - 29, 2023, San Francisco, CA.
- Ghosh, D.**, Dorr, M., Dorf, M., Ricketson, L., *Semi-Implicit Time Integration for Multiscale Tokamak-Edge Plasma Dynamics*, SIAM Conference on Computational Science and Engineering, Minisymposium on Computational Methods for Kinetic Models, February 26 - March 3, 2023, Amsterdam, The Netherlands.
- Bonneville, C., Choi, Y., **Ghosh, D.**, *GPLaSDI: Identifying Autoencoders Latent Space Dynamics with Gaussian Processes for Non-Intrusive Reduced Order Models*, SIAM Conference on Computational Science and Engineering, Minisymposium on Reduced Order Modeling of Differential Equations Through Deep Learning Algorithms, February 26 - March 3, 2023, Amsterdam, The Netherlands.
- High-Order Semi-Implicit Time Integration Methods for Multiscale Advective Physics*, University of Zaragoza, October 2022 (Online talk).
- Ricketson, L., Dorr, M., Dorf M., **Ghosh, D.**, *Continuum Kinetic Simulations of Tokamak Edge Plasmas with COGENT*, Platform for Advanced Scientific Computing (PASC) Conference, Minisymposium on High Performance Computing in Kinetic Simulations of Plasmas - Part III: Advanced Numerical Methods and Algorithms, June 27 - 29, 2022, Basel, Switzerland.
- Ricketson, L., **Ghosh, D.**, Dorr, M., Dorf M., *Block Preconditioning of a Semi-Implicit Gyrokinetic Model of Fusion Plasmas*, 17th Copper Mountain Conference On Iterative Methods, Session on Scalable Solvers for Coupled Multiphysics 2, April 4 - 8, 2022, Virtual Format.
- Ricketson, L., **Ghosh, D.**, Dorr, M., Dorf M., Hittinger, J., *Continuum Kinetic Simulations of Tokamak Edge Plasmas with COGENT*, Platform for Advanced Scientific Computing (PASC) Conference, Minisymposium on Towards Exascale Computing in Kinetic Simulations of Magnetic Fusion Plasmas. Part III - Advanced Numerical Methods and Algorithms, July 5 - 9, 2021, Virtual Format.
- Ricketson, L., **Ghosh, D.**, Dorr, M., Dorf M., *Block Preconditioning of a Semi-Implicit Gyrokinetic Model of Fusion Plasmas*, 9th International Conference on Computational Methods for Coupled Problems in Science and Engineering (COUPLED PROBLEMS 2021), Minisymposium on Block Preconditioning for Challenging Multiphysics Systems, June 14 - 16, 2021, Virtual Format.
- Ghosh, D.**, Chapman, T., Berger, R. L., Banks, J. W., Copeland, D., *A Semi-Implicit Algorithm for the Simulation of High-Z Plasma Interpenetration*, 15th United States National Congress on Computational Mechanics (USNCCM15), Minisymposium on Innovative Higher Order Discretization Methods in Computational Science and Engineering, in honor of Prof. Mary Wheeler, July 28 - August 1, 2019, Austin, TX.
- Hittinger, J., Dorr, M. R., Dorf, M. A., **Ghosh, D.**, Ricketson, L. F., *Kinetic Edge Plasma Simulation using a Continuum Model*, Platform for Advanced Scientific Computing (PASC), Minisymposium on HPC Challenges in Kinetic Simulations of Plasmas, Part I of III: Eulerian Approach, June 12 - 14, 2019, Zurich, Switzerland.
- Ghosh, D.**, Chapman, T. D., Berger, R. L., Dimits, A., Banks, J. W., *A Multi-Species, Multi-Fluid Model for Simulating Plasma Interpenetration*, SIAM Conference on Computational Science and Engineering, Minisymposium on State-of-the-art high-order Numerical Methods and Complex Fluid Simulations, February 25 - March 1, 2019, Spokane, WA.
- Banks, J. W., Berger, R. L., Brunner, S., Chapman, T. D., Dimits, A., **Ghosh, D.**, *Simulation of Interpenetrating Plasmas in 2D using the Grid Based Continuum Code LOKI*, SIAM Conference on Computational Science and Engineering, Minisymposium on Computational Methods for Kinetic Models of Plasma, February 25 - March 1, 2019, Spokane, WA.
- Ghosh, D.**, Kavouklis, C., Chapman, T., Berger, R. L., Vogman, G. V., *Numerical Simulation of Counterstreaming Plasma Interactions using a Multifluid Model*, 13th World Congress on Computational Mechanics (WCCM XIII), Minisymposium on High-order Discretizations for Multi-physics Applications, July 22 - 27, 2018, New York, NY.
- Dimits, A. M., Berger, R. L., Chapman, T., **Ghosh, D.**, Arrighi W. J., Hittinger, J. A. F., Joseph, I., Kavouklis, C., Brunner, S., Banks, J. W., *Linearized Coulomb Collision Operator for Simulation of Interpenetrating Plasma Streams*, 45th IEEE International Conference on Plasma Science, June 24 - 28, 2018, Denver, CO.
- Implicit-Explicit Time Integration for Multiscale Physics*, Department of Applied Mathematics and Statistics, University of California Santa Cruz, December 2017, Santa Cruz, CA.
- Ghosh, D.**, Constantinescu, E. M., *Characteristic-Based Slow-Wave-Fast-Wave Partitioning for Semi-Implicit Time Integration of Atmospheric Flows*, IMAGE 2017 Theme of the Year: Workshop on Multiscale Geoscience Numerics, National Center for Atmospheric Research (NCAR), May 16 - 19, 2017, Boulder, CO.
- Ghosh, D.**, Dorr, M., Dorf, M., Hittinger, J., *Implementation of Implicit-Explicit Time Integration for the Kinetic Modeling of Tokamak Plasma Edge*, SIAM Conference on Computational Science and Engineering, Minisymposium on Computational Plasma Physics, February 27 - March 3, 2017, Atlanta, GA.
- Dorr, M., Colella, P., Dorf, M., **Ghosh, D.**, Hittinger, J., *Exact Preservation of Zero Velocity Divergence in a High-Order, Mapped-Grid, Finite-Volume Discretization of a Gyrokinetic System*, SIAM Conference on Computational Science and Engineering, Minisymposium on Computational Plasma Physics, February 27 - March 3, 2017, Atlanta, GA.
- Ghosh, D.**, Constantinescu, E. M., *Characteristic-Based Flux Splitting for Implicit-Explicit Time Integration of Low-Mach Number Flows*, 13th U.S. National Congress on Computational Mechanics (USNCCM13), Minisymposium on Advances in Implicit / Explicit (IMEX) Time integration of Multiphysics Systems, July 26 - 30, 2015, San Diego, CA.
- Ghosh, D.**, Constantinescu, E. M., *A Finite-Difference Algorithm with Characteristic-Based Semi-Implicit Time-Integration for the Euler*

Equations with Gravitational Forcing, SIAM Conference on Mathematical & Computational Issues in the Geosciences, Minisymposium on Modeling and Simulation of Multiscale and Coupled Processes in Atmospheric Physics, June 29 - July 2, 2015, Stanford, CA.

Ghosh, D., Constantinescu, E. M., *A Compact-Reconstruction WENO Scheme with Semi-Implicit Time Integration*, SIAM Conference on Computational Science and Engineering, Minisymposium on Recent Advances in High Order Spatial Discretization Methods for PDEs, March 14 - 18, 2015, Salt Lake City, UT.

Ketcheson, D., **Ghosh, D.**, *Stability-Optimized Time Integrators for WENO Discretizations*, SIAM Conference on Computational Science and Engineering, Minisymposium on Advances in Time-stepping Methods, March 14 - 18, 2015, Salt Lake City, UT.

Barajas-Solano, D. A., Tarakovsky, A., **Ghosh, D.**, Constantinescu, E. M., Abhyankar, S., *Probability Density Methods for the Analysis of Power Grids Under Uncertainty*, SIAM Conference on Computational Science and Engineering, Minisymposium on Distributed Cyber-Physical Systems: Modelling and Controlling the Power Grid, March 14 - 18, 2015, Salt Lake City, UT.

Ghosh, D., Constantinescu, E. M., Brown, J., *Scalable Non-Linear Compact Schemes*, International Conference on Spectral and High Order Methods (ICOSAHOM), Minisymposium on Aspects of Time Stepping, June 23 - 27, 2014, Salt Lake City, UT.

Compact-Reconstruction WENO Schemes - Theory, Implementation and Applications, National Institute of Aerospace, June 2014, Hampton, VA.

OTHER CONFERENCES, MEETINGS, TALKS AND POSTERS (55)

Angus, J. R., Grote, D., **Ghosh, D.**, van de Wetering, J. J., Kemp, A., Vay, J.-L., *Energy-Conserving Implicit Particle-in-Cell Method in WarpX for Dense, Fusion Plasmas*, 2026 IEEE International Conference on Plasma Science (ICOPS), 93.

Lehe, R., Angus, J. R., **Ghosh, D.**, Groenewald, R., Huebl, A., Formenti, A., Grote, D., Myers, A., Shalpoval, O., Vay, J.-L., Zhang, W., Zoni, E., *WarpX for Fusion Research: Recent Advances and New Capabilities*, 67th Annual Meeting of the American Physical Society's Division of Plasma Physics (DPP), November 17 - 21, 2025, Long Beach, CA.

Oudin, A., **Ghosh, D.**, Kur, E., Lancia, L., Riconda, C., Ou, K., Perez-Ramirez, V. M., Cao, S., Singh, D., Redshaw, C., Rajesh, H., Dedeler, P., Edwards, M. R., Michel, P., *Nonlinear Modeling of Photochemically-Induced Gaseous Optical elements*, 13ème Forum Lasers et Plasmas, June 2 - 6, 2025, Le Plan-de-la-Tour, France. Also presented at: 53rd Annual Anomalous Absorption Conference, May 11 - 16, 2025, Sedona, AZ.

Dorf, M., Dorr, M., **Ghosh, D.**, Umansky, M., Geyko, V., Angus, J., Soukhanovskii, V., *Implicit simulations of collisional transport in magnetized plasmas with the continuum gyrokinetic code COGENT*, Sherwood Fusion Theory Conference, April 7 - 9, 2025, New York, NY.

Angus, J. R., van de Wetering, J. J., Farmer, W., Friedman, A., Fu, Y., Geyko, V., **Ghosh, D.**, Grote, D., Larson, D., Zimmerman, G., *An implicit particle code for fully kinetic simulations of burning plasmas*, IFE-STAR Conference, April 7, 2025, Breckenridge, CO.

Terasaki, G. M., **Ghosh, D.**, Navid, A., *Examining Complex Disease Spread Dynamics in Schools and Workplaces Using An Exascale Agent-Based Framework*, SIAM Conference on Computational Science and Engineering, March 3 - 7, 2025, Fort Worth, TX.

Salian, L. V., Rathan, S., **Ghosh, D.**, *A new central compact finite difference scheme with high spectral resolution for dispersive equations*, 19th International Conference on Hyperbolic Problems: Theory, Numerics and Applications, July 1 - 5, 2024, Shanghai, China.

Ricketson, L., Dorr, M., Dorf, M., **Ghosh, D.**, *Solvers and preconditioners enabling flexible IMEX integration of fusion plasmas in COGENT*, 18th Copper Mountain Conference on Iterative Methods, April 14 - 19, 2024, Copper Mountain, CO.

Angus, J., Farmer, W., Friedman, A., Geyko, V., **Ghosh, D.**, Grote, D., Larson, D., Link, A., *An implicit particle code with exact energy and charge conservation for dynamically compressing plasmas: Solver scaling and performance*, 18th Copper Mountain Conference on Iterative Methods, April 14 - 19, 2024, Copper Mountain, CO.

Dorf, M., Dorr, M., **Ghosh, D.**, Angus, J. R., *Modeling tokamak edge plasma with the continuum full-F code COGENT*, 65th Annual Meeting of the American Physical Society's Division of Plasma Physics (DPP), October 30 - November 3, 2023, Denver, CO.

Dun, J., Angus, J. R., Farmer, W., Friedman, A., Geyko, V. I., **Ghosh, D.**, Graziani, F. R., Grote, D., Larson, D., Link, A. J., Zimmerman, G. B., *Particle-in-cell simulations of shockwaves in plasma*, 65th Annual Meeting of the American Physical Society's Division of Plasma Physics (DPP), October 30 - November 3, 2023, Denver, CO.

Angus, J. R., Farmer, W., Friedman, A., Ghosh, D., Grote, D., Larson, D., Link, A. J., *An implicit particle code with exact energy and charge conservation for electromagnetic studies of dense plasmas*, 65th Annual Meeting of the American Physical Society's Division of Plasma Physics (DPP), October 30 - November 3, 2023, Denver, CO.

Tsai, P.-H., Chung, S. W., **Ghosh, D.**, Choi, Y., Belof, J., *Accelerating Kinetic Simulations of Electrostatic Plasmas with Reduced-Order Modeling*, 65th Annual Meeting of the American Physical Society's Division of Plasma Physics (DPP), October 30 - November 3, 2023, Denver, CO.

Ghosh, D., Dorr, M., Dorf, M., Ricketson, L., *High-order implicit-explicit time integration for the kinetic simulation of magnetized plasmas*, Foundations of Computational Mathematics (FoCM), Workshop on Multiresolution and Adaptivity in Numerical PDEs, June 12 - 21, 2023, Paris, France.

Angus, J., Farmer, W., Friedman, A., **Ghosh, D.**, Grote, D., Larson, D., Link, A., *An implicit particle code with exact energy and charge*

- conservation for electromagnetic studies of dense plasmas, International Conference on Plasma Science (ICOPS), May 21 - 25, 2023, Santa Fe, NM.
- Dorf, M., Dorr, M., **Ghosh, D.**, Angus, J., *Including electromagnetic effects into COGENT simulations of edge plasmas*, U.S. Transport Task Force (TTF) Workshop, May 2 - 5, 2023, Madison, WI.
- Geyko, V., Joseph, I., Dorf, M., Dorr, M., **Ghosh, D.**, *Electrostatic drift-kinetic simulations of edge localized mode heat pulses in the tokamak scrape-off layer*, 64th Annual Meeting of the American Physical Society's Division of Plasma Physics (DPP), October 17 - 21, 2022, Spokane, WA.
- Dorf, M., Dorr, M., **Ghosh, D.**, Angus, J., *Progress with the full-F continuum edge gyrokinetic code COGENT*, 64th Annual Meeting of the American Physical Society's Division of Plasma Physics (DPP), October 17 - 21, 2022, Spokane, WA.
- Geyko, V., Joseph, I., Dorf, M., Dorr, M., **Ghosh, D.**, *Drift kinetic simulations of edge localized modes in tokamak scrape-off layer plasmas*, 25th International Conference on Plasma Surface Interaction in Controlled Fusion Devices, June 13 - 17, 2022, Virtual Format.
- Angus, J. R., Link, A., Friedman, A., **Ghosh, D.**, Johnson, J. D., *On numerical energy conservation for an implicit particle-in-cell method coupled with a binary Monte-Carlo algorithm for Coulomb collisions*, 49th IEEE International Conference on Plasma Science (ICOPS), May 22 - 26, 2022, Seattle, WA.
- Dorf, M., Dorr, M., **Ghosh, D.**, Cohen, R., *Gyrokinetic Modeling of Electrostatic Ion-Scale Turbulence in Divertor Tokamaks*, 63rd Annual Meeting of the American Physical Society's Division of Plasma Physics (DPP), November 8 - 12, 2021, Pittsburgh, PA.
- Joseph, I., Geyko, V. I., Dorf, M. A., Dorr, M. R., **Ghosh, D.**, Ricketson, L. F., *Edge localized mode propagation through the tokamak scrape-off layer*, Plasma Edge Theory (PET) Workshop, September 13 - 15, 2021, Virtual Format.
- Ghosh, D.**, Berger, R. L., Chapman, T., Dimits, A., Banks, J. W., *Hydrodynamic Simulation of Counterstreaming Plasmas with a Multifluid Model*, 72nd Annual Meeting of the American Physical Society's Division of Fluid Dynamics (DFD), November 23 - 26, 2019, Seattle, WA.
- Chapman, T., Berger, R. L., Dimits, A., **Ghosh, D.**, Joseph, I., Winjum, B., Banks, J. W., Brunner, S., *Nonlinear evolution of the ion-ion streaming instability in single- and multi-ion species plasmas*, 61st Annual Meeting of the APS Division of Plasma Physics (DPP), October 21 - 25, 2019, Fort Lauderdale, FL.
- Berger, R., **Ghosh, D.**, Banks, J. W., Arrighi, W. J., Belyaev, M. A., Brunner, S., Chapman, T., Copeland, D. M., Dimits, A., Hittinger, J., Jones, O., Rozmus, W., *Multi-dimensional Simulations of Interpenetrating Plasmas*, 49th Anomalous Absorption Conference, June 9 - 14, 2019, Telluride, CO.
- Berger, R., **Ghosh, D.**, Chapman, T., Arrighi, W., Banks, J., Dimits, A. M., Hittinger, J., Joseph, I., Kavouklis, C., *High-order Conservative, Eulerian, Multi-dimensional Hydrodynamic Simulations of Interpenetrating Plasmas*, 60th Annual Meeting of the APS Division of Plasma Physics, November 5 - 9, 2018, Portland, OR.
- Angus, J. R., Dorf, M., Ghosh D., *Drift-Ideal MHD Simulations of Flow-Stabilized Z-Pinch Plasmas*, 60th Annual Meeting of the APS Division of Plasma Physics, November 5 - 9, 2018, Portland, OR.
- Ghosh, D.**, Kavouklis, C., Chapman, T., Berger, R., *A Finite-Volume, Semi-Implicit Multifluid Algorithm for the Simulation of Counterstreaming Plasma Dynamics*, SIAM Annual Meeting, July 9 - 13, 2018, Portland, OR.
- Dorf, M. A., Angus, J. R., **Ghosh, D.**, *Gyrokinetic Simulations of Drift-Wave Instabilities in Flow-Stabilized Z-Pinch Plasmas*, 45th IEEE International Conference on Plasma Science, June 24 - 28, 2018, Denver, CO.
- Angus, J. R., Dorf, M. A., **Ghosh, D.**, *Drift-Ideal MHD Simulations of the Entropy Mode in Flow-Stabilized Z-Pinch Plasmas*, 45th IEEE International Conference on Plasma Science, June 24 - 28, 2018, Denver, CO.
- Dimits, A. M., Banks, J. W., Berger, R. L., Brunner, S., Chapman, T., **Ghosh, D.**, Arrighi, W. J., Hittinger, J., Joseph, I., Kavouklis, C., *Linearized Coulomb Collision Operator for Simulation of Interpenetrating Plasma Streams*, Kinetic Effects in ICF Workshop, May 23, 2018, Santa Fe, NM.
- Dorf, M., Dorr, M., **Ghosh, D.**, Hittinger, J., Lee, W., Cohen R., *Drift-kinetic simulations of axisymmetric plasma transport at the edge of a divertor tokamak*, 59th Annual Meeting of the APS Division of Plasma Physics, October 23 - 27, 2017, Milwaukee, WI.
- Ghosh, D.**, Kavouklis, C., Berger, R. L., Chapman, T., Hittinger, J., *A Multifluid Numerical Algorithm for Interpenetrating Plasma Dynamics*, 59th Annual Meeting of the APS Division of Plasma Physics, October 23 - 27, 2017, Milwaukee, WI.
- Ghosh, D.**, Chapman, T., Berger, R. L., Khodak, M., Hittinger, J., *Simulation of interpenetrating plasmas in 1D with a multifluid approach*, 47th Annual Anomalous Absorption Conference, June 11 - 16, 2017, Florence, OR.
- Lee, W., Dorf, M., Dorr, M., Cohen, R., **Ghosh, D.**, Hittinger, J., Umansky, M., Krashenninnikov, S., *Status of 5D continuum kinetic code COGENT and its verification with kinetic drift wave instability*, International Sherwood Fusion Theory Conference, May 1 - 3, 2017, Annapolis, MD.
- Dorf, M., Dorr, M., **Ghosh, D.**, Hittinger, J., Lee, W., Cohen, R., *Continuum kinetic modeling of cross-separatrix plasma transport with COGENT*, International Sherwood Fusion Theory Conference, May 1 - 3, 2017, Annapolis, MD.
- Kavouklis, C., Colella, P., **Ghosh, D.**, *A 6th order finite volume multigrid Poisson solver*, 15th Copper Mountain Conference on Iterative

Methods, March 25 - 30, 2017, Copper Mountain, CO.

- Lee, W., Dorf, M., Dorr, M., Cohen, R., **Ghosh, D.**, Rognlien, T., Hittinger, J., Umansky, M., Krasheninnikov, S., *Status of the 5D gyrokinetic code COGENT and its initial applications*, 58th Annual Meeting of the APS Division of Plasma Physics, October 31 - November 4, 2016, San Jose, CA.
- Dorf, M., Dorr, M., **Ghosh, D.**, Hittinger, J., Rognlien, T., Cohen, R., Lee, W., Schwartz, P., *Edge-relevant plasma simulations with the continuum code COGENT*, 58th Annual Meeting of the APS Division of Plasma Physics, October 31 - November 4, 2016, San Jose, CA.
- Dorf, M., Dorr, M., Cohen, R. H., Lee, W., **Ghosh, D.**, Hittinger, J., Rognlien, T., *Continuum Kinetic Modeling of the Tokamak Plasma Edge with the COGENT code*, International Sherwood Fusion Theory Conference, April 4 - 6, 2016, Madison, WI.
- Cohen, R. H., Dorf, M., Dorr, M., Lee, W., **Ghosh, D.**, Hittinger, J., Rognlien, T., *Continuum Kinetic Modeling of the Tokamak Plasma Edge with the COGENT*, US Transport Task Force Workshop, March 29 - April 1, 2016, Denver, CO.
- Ghosh, D.**, Constantinescu, E. M., Brown, J., *A Scalable, Parallel Implementation of Weighted, Non-Linear Compact Schemes*, SIAM Annual Meeting, July 7 - 11, 2014, Chicago, IL.
- Brown, J., **Ghosh, D.**, *Fast solvers for implicit Runge-Kutta*, Thirteenth Copper Mountain Conference on Iterative Methods, April 6 - 11, 2014, Copper Mountain, CO.
- Brown, J., **Ghosh, D.**, *Fast solvers for implicit Runge-Kutta systems*, Tenth Workshop of the INRIA-Illinois-ANL Joint Laboratory on Petascale Computing, November 25 - 27, 2013, Urbana, IL.
- Brown, J., Constantinescu, E. M., **Ghosh, D.**, McInnes, L.C., *BOUT++ and PETSc: Time integration and solver composition*, 2013 BOUT++ Workshop, September 3 - 6, 2013, Lawrence Livermore National Laboratory, CA.
- Constantinescu, E. M., Brown, J., Smith, B., McInnes, L.C., **Ghosh, D.**, Balay, S., *Robust Implicit-Explicit Time Stepping*, DOE Applied Math PI Meeting, August 6, 2013, Albuquerque, NM.
- Ghosh, D.**, Medida, S., Baeder, J. D., *Direct Numerical Simulation of Compressible Turbulent Flows with Weighted Non-Linear Compact Schemes*, 65th Annual Meeting of the APS Division of Fluid Dynamics, November 18 - 20, 2012, San Diego, CA.
- Ghosh, D.**, Medida, S., Baeder, J. D., *High-Order Non-Oscillatory Compact Reconstruction Scheme for Overset Grids*, 11th Symposium on Overset Composite Grids and Solution Technology, October 15 - 18, 2012, Dayton, OH.
- Ghosh, D.**, Baeder, J. D., *Application of Compact-Reconstruction WENO Schemes to the Navier-Stokes Equations*, SIAM Annual Meeting, July 9 - 13, 2012, Minneapolis, MN.
- Ghosh, D.**, Baeder, J. D., *High Order Compact Reconstruction Scheme with Weighted Essentially Non-Oscillatory Limiting*, AMS-MAA Joint Mathematics Meetings, Jan 4-7, 2012, Boston, MA.
- Ghosh, D.**, *Application of Weighted Essentially Non-Oscillatory Limiting to Compact Interpolation Schemes*, AIAA Region I Young Professional, Student & Education Conference, Nov 4, 2011, Laurel, MD.
- Ghosh, D.**, Chatterjee, A., *Higher-Order Non-Oscillatory Schemes in 2D Ideal Magneto-hydrodynamics*, 9th Annual CFD Symposium, August 2006, Bangalore.
- Ghosh, D.**, Chatterjee, A., *Higher-Order Non-Oscillatory Schemes in Ideal Magneto-hydrodynamics*, 8th Annual CFD symposium, August 2005, Bangalore.
- Ghosh, D.**, Chatterjee, A., *Radar Cross Section Computations for Low Observable Configurations*, 18th National Conference for Aerospace Engineers, Institution of Engineers (I), November, 2004, IIT Kharagpur.
- Ghosh, D.**, Vaghela, M.B., Chatterjee, A., *Characteristic Based FVTD Computations for Low Observable Configuration*, 7th Annual CFD symposium, August 2004, Bangalore.

RESEARCH FUNDING

- Addressing key physics problems in high-energy-density plasmas with a novel kinetic simulation capability (**Co-Investigator**) — LLNL Laboratory Directed Research and Development (LDRD), ~\$650k/yr, 2023–2026.
- High-resolution methods for phase space problems in complex geometries (**Co-Investigator**) — DOE Office of Science, ASCR, ~\$900k/yr, 2017–2020 and 2021–2024.
- Simulation of high-energy-density plasmas using a novel, fully implicit particle-in-cell / Monte-Carlo algorithm (**Co-Investigator**) — LLNL LDRD, ~\$150k/yr, 2021–2022.
- Nonlinear spatial discretization on sparse grids (**Co-Investigator**) — LLNL LDRD, ~\$150k/yr, 2019–2020.
- Interpenetrating plasma simulations (**Principal Investigator**) — LLNL LDRD, ~\$650k/yr, 2017–2020.

PROFESSIONAL SERVICE

- Member, AIAA Atmospheric and Space Environments Technical Committee, 2016 – present.
- **Reviewer** — SIAM Journal on Scientific Computing; Journal of Computational Physics; Journal of Scientific Computing;

ACM Transactions on Mathematical Software; Computer Physics Communications; Communications in Computational Physics; Physics of Fluids; Journal of Parallel and Distributed Computing; Journal of Advances in Modeling Earth Systems; International Journal for Numerical Methods in Fluids; Computers and Mathematics with Applications; International Journal of Computational Fluid Dynamics; International Journal of High Performance Computing Applications; International Journal of Computer Mathematics; SC17.

- **Session chair / co-chair** — 7th AIAA Atmospheric and Space Environments Conference (Numerical Weather Prediction, with Nash'at Ahmad); SIAM Annual Meeting 2014 (Numerical Methods in PDE VII).
- **Visiting researcher** — Department of Applied Mathematics, Naval Postgraduate School (host: Frank Giraldo), September 2015; Computer, Electrical and Mathematical Sciences and Engineering, King Abdullah University of Science and Technology (host: David Ketcheson), June 2015.
- **Organizer** — LANS Informal Seminar Series, Mathematics and Computer Science Division, Argonne National Laboratory, 2013–2015.

HONORS AND AWARDS

- Travel award, International Conference on Spectral and High Order Methods, 2014.
- Graduate Research Assistantship, Alfred Gessow Rotorcraft Center, University of Maryland, 2008–2013.
- Block Grant Fellowship, Department of Mathematics, University of Maryland, 2006–2008.
- Research Assistantship, Department of Aerospace Engineering, Indian Institute of Technology Bombay, 2005–2006.

TEACHING

Teaching Assistant

Jul 2005 – May 2006

Department of Aerospace Engineering, Indian Institute of Technology Bombay

- Numerical Methods for Conservation Laws (graduate) — lectured on finite-volume methods for ideal magnetohydrodynamics; graded coding assignments.
- Compressible Gasdynamics (undergraduate) — graded assignments and examinations.